

Adaptive Evacuation Route Planning under Crowded and Dynamic Fire Conditions

An agent-based simulation study of routing algorithms and decision support under limited visibility

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Abstract

We study crowd evacuation in a multi-room building under a dynamic multi-source fire, where 50% of occupants are unfamiliar with the building and do not know the globally optimal exit. Occupants move with a Social Force Model, are driven by a dynamic fire-hazard field and a contagion-based fear model, and route by minimizing a composite cost of time, hazard, congestion, and fear. We compare Dijkstra, A*, ACO, and RL-ACO, and dissect the decision-support layer (static signs + dynamic guides) with controlled ablations.

Three findings emerge. (1) RL-ACO - an A*-based router with mobile, hazard-intercepting guides - reaches a trimmed-median 95%-evacuation time (T95) of 125.3 s versus 205.9 s (Dijkstra), 255.5 s (A*), and 476.1 s (ACO), with the lowest exposure (12.9) and fewest trapped (5). (2) The guide layer matters only when guides can MOVE and INTERCEPT: stationary guides (fixed/random/ACO exit placement) are indistinguishable, while mobile interception adds a further 31.2% over random. (3) Static, fire-unaware signage is a double-edged sword - it can steer people toward fire and raise exposure.

Key results (24 runs / variant, trimmed-median)		
RL-ACO	125.3 s	T95 - 39% faster than Dijkstra, 51% than A*, 74% than ACO
Dijkstra / A* / ACO	205.9 / 255.5 / 476.1 s	baselines (ACO fails to converge on multi-room map)
Mobile vs random guides	-31.2% T95	exposure 12.9 vs 31.6; trapped 5 vs 69
Static signs	harmful	raise exposure 24.5 -> 28.0 and trapped 11 -> 17

Methodology in brief: 60 x 40 m multi-room floor (6 exits); 100 occupants, 50% unfamiliar; 3-source fast t^2 fire; Social Force Model motion + Weidmann density-speed + fear/panic (wandering, deviating, rushing, herding); dynamic edge cost (time + hazard + congestion + fear); guide risk score = $2 \times \text{panic} + \text{unfamiliar} \times (1 + 3 \times \text{exit-hazard})$. Full details and the complete experimental history (including negative results) are in the accompanying REPORT.md.

2. Model & Experimental Setup

- Scene	60x40 m, 3 halls separated by walls with 8 m doorways; 4 columns; seating zones; 6 exits (Main 6 m, others 3 m).
- Motion	Social Force Model (Helbing 2000): relaxation $\tau=0.5$ s, pedestrian repulsion $A=2000$ N, wall repulsion $A_w=1000$ N; Weidmann density-speed; slow zones across seat-row gaps.
- Fire	3 sources, t^2 growth ($\alpha=0.0469$ kW/s ² , fast), saturate 40 kW, radius 1-10 m; Gaussian plume (heat/smoke/CO/invisibility) $\rightarrow$ scalar $h_{in}[0,1]$; router sees h delayed 10 s + 5% noise.
- Fear & panic	$df/dt = k_H \cdot H + k_D \cdot \rho + k_C \cdot \text{contagion} + k_U \cdot (1 - \text{familiarity}) - \text{decay} \cdot f$; non-monotonic speed; above threshold: herding / wandering / deviating / rushing.
- Limited visibility	unfamiliarFrac = 0.5; unfamiliar agents know one random exit, navigate the graph to it only; learn better exits from signs/guides.
- Routing	Graph (grid + aisle-centre + row-gap waypoints, LoS edges); cost = $w_t \cdot \text{time} + w_h \cdot \text{hazard} + w_c \cdot \text{congestion} + w_f \cdot \text{fear}$ ($w = 1.0/2.5/1.5/1.2$) + high-hazard penalty.
- Guides	2 mobile marshals; radius 8 m; speed 2 m/s; risk = $2 \cdot x_{\text{panic}} + \text{unfamiliar} \cdot (1 + 3 \cdot x_{\text{hazard at target exit}})$; re-target every 4 s, move every step along the graph.
- Protocol	$dt=0.05$ s; replan 1 s (global) / 2 s (agent); 600 s cap; 12 runs x 2 batches = 24 runs/variant; trimmed median (drop fastest & slowest).

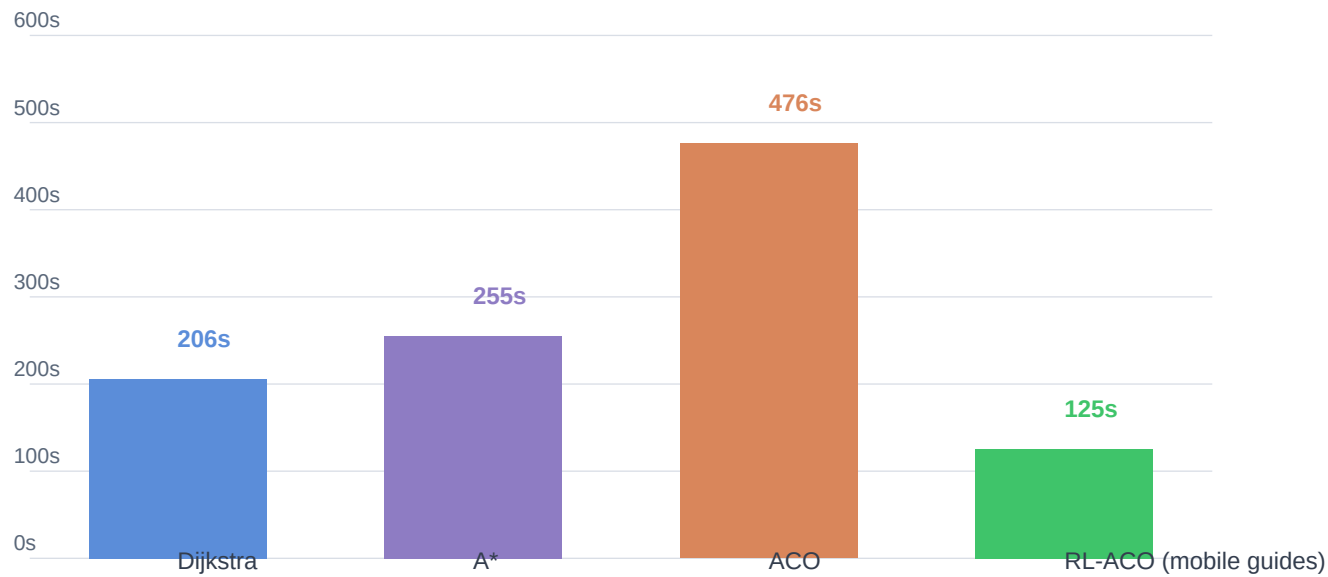
Metrics

Primary: T95 - time until 95% of occupants have evacuated (null if not reached by 600 s). Secondary: number of trapped (non-evacuated) agents, and mean hazard exposure (cumulative true hazard). We also record evacuation-progress, mean-fear, peak-density time series, and per-exit usage.

3. Algorithm Comparison

T95 (trimmed median, 24 runs)

All methods share static signs; RL-ACO additionally deploys mobile hazard-intercept guides.



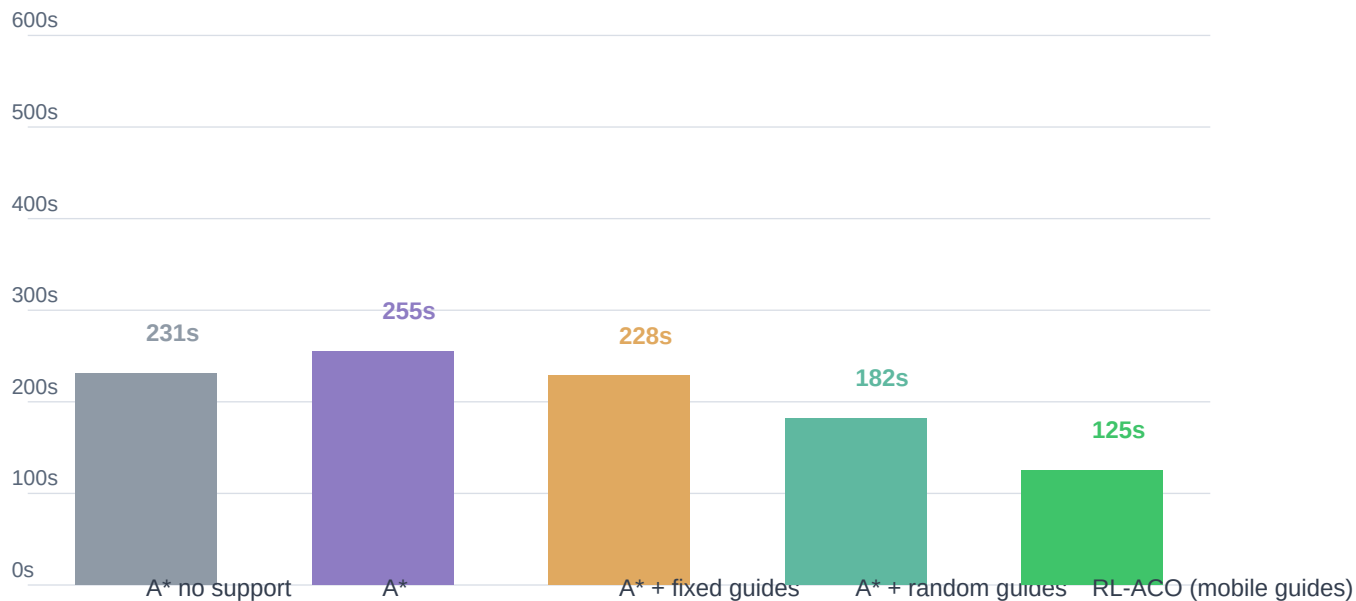
Algorithm	T95 median	T95 mean	Trapped	Exposure
Dijkstra	205.9 s	213.1 s	11	23.1
A*	255.5 s	253.4 s	17	28.0
ACO	476.1 s	478.0 s	93	50.7
RL-ACO (mobile guides)	125.3 s	133.7 s	5	12.9

RL-ACO is 39% faster than Dijkstra, 51% faster than A*, and 74% faster than ACO, and simultaneously achieves the lowest exposure and fewest trapped occupants. ACO (ant pheromone routing) fails to converge on the multi-room topology: 7 of 24 runs time out.

4. Decision-Support Ablation (fixed A* routing)

T95 (trimmed median, 24 runs)

Adding support layer by layer isolates exactly what helps.



Variant	T95 median	Trapped	Exposure
A* no support	231.1 s	11	24.5
A*	255.5 s	17	28.0
A* + fixed guides	228.5 s	62	35.2
A* + random guides	182.2 s	69	31.6
RL-ACO (mobile guides)	125.3 s	5	12.9

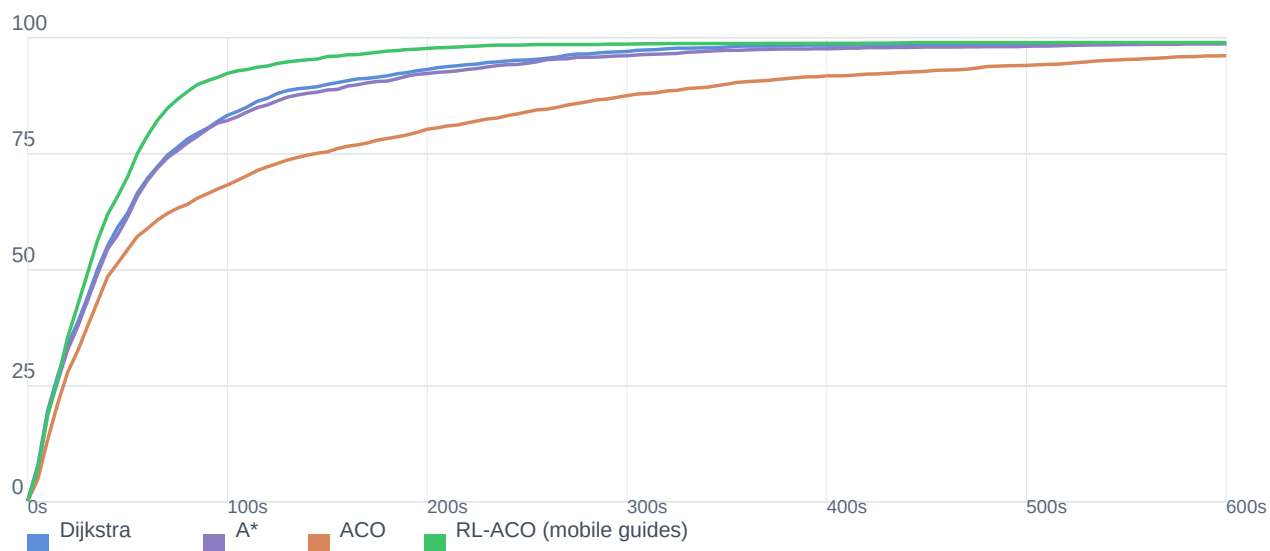
- Static signs (2 vs 1): T95 up 231.1 -> 255.5 s, exposure 24.5 -> 28.0, trapped 11 -> 17 - fire-unaware sign ACO (multiple guides) toward bottom exits near fire.

- Fixed / random guides (3,4 vs 1): modest speedup but WORSE exposure and trapping - stationary guides pull people toward (possibly dangerous) exits; placement among exits is indistinguishable.

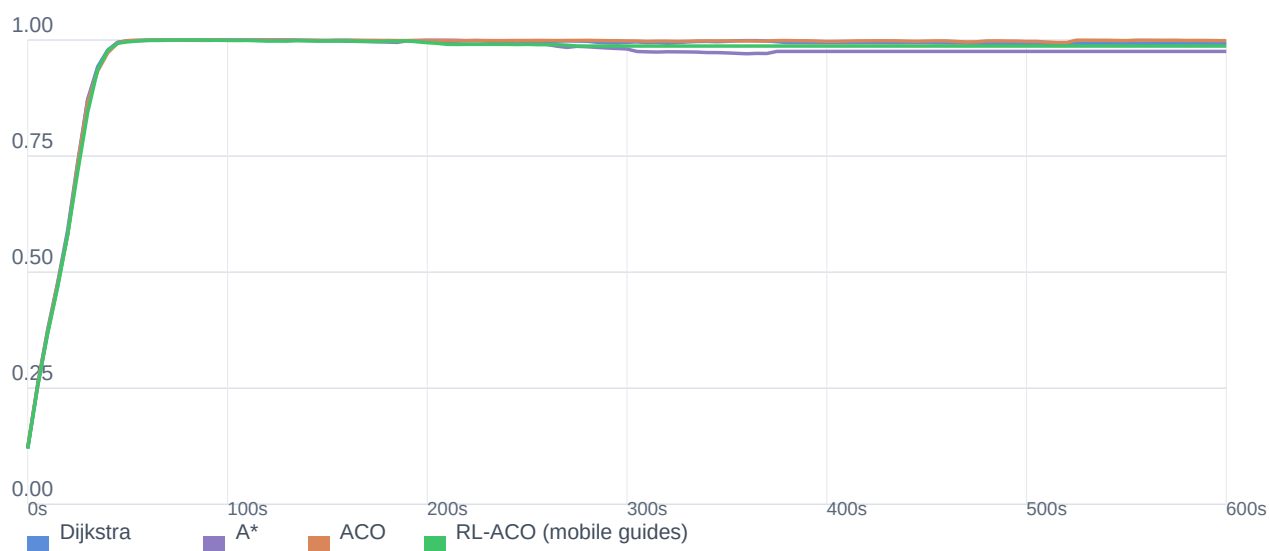
- Mobile hazard-intercept guides (5 vs 4): T95 182.2 -> 125.3 s (-31.2%); exposure 31.6 -> 12.9; trapped 69 -> 5 - the only mechanism that improves all three metrics at once.

5. Time Series (mean over 24 runs)

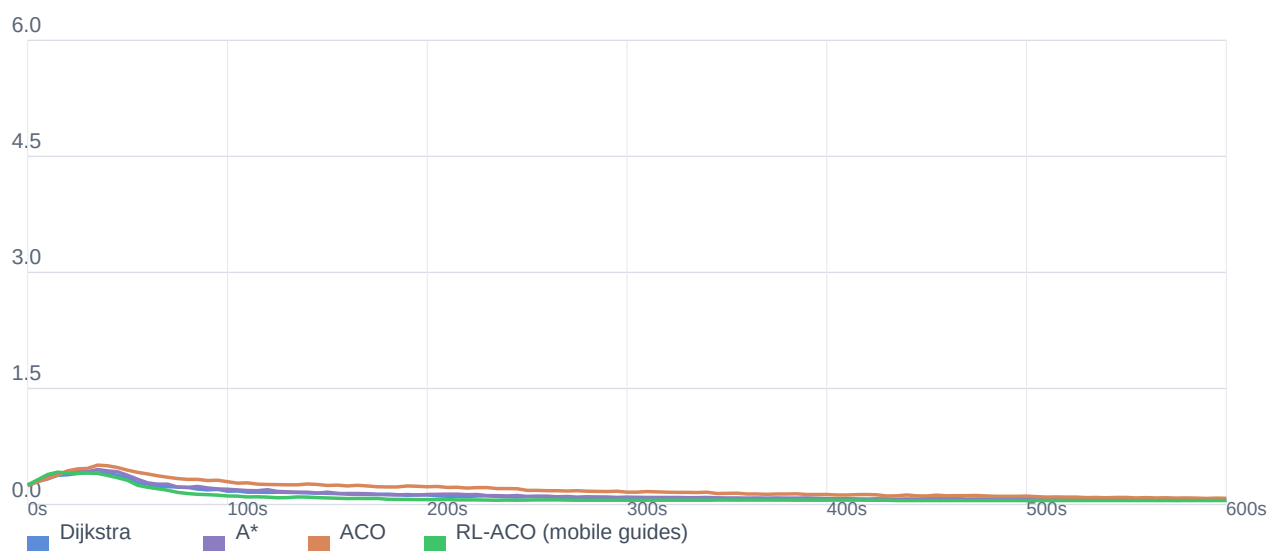
Evacuation progress (%)



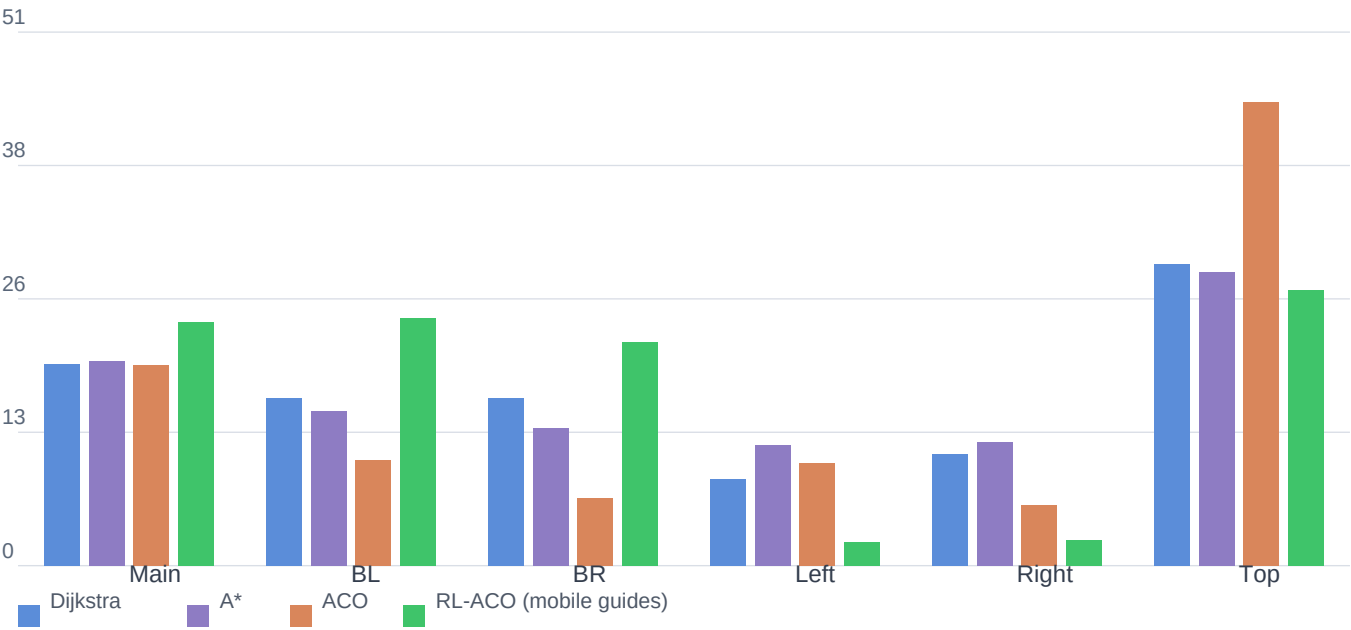
Mean fear (0-1)



Peak density (persons/m^2)



6. Exit Usage & Robustness



Robustness: mobile vs random guides across independent experiments

Experiment (runs)	Mobile T95	Random T95	Mobile
E1 (12 runs)	124.2 s	193.0 s	-35.6%
E2 (24 runs)	128.4 s	179.6 s	-28.5%
E3 (24 runs, this report)	125.3 s	182.2 s	-31.2%

The mobile-guide advantage never reverses across the three independent experiments (single-batch magnitudes -18.5% to -35.6%). The earlier "adaptive ~ random" null result applied only to EXIT-BASED (stationary) placement; mobility + hazard interception is what makes the guide layer genuinely effective.

7. Discussion & Conclusion

- Exit-based placement does not matter.

Every stationary guide teaches the same optimal exit ($\text{recExit} = \text{bestExit}[\text{node}]$) to whoever passes within 8 m; placement only changes coverage, and unfamiliar occupants are spread over all six exits. Adaptive-vs-random differences flipped sign (-22%, +4%, -2%) - pure noise.

- Mobility + hazard interception is the real mechanism.

A mobile guide is assigned each 4 s to the highest-risk occupant(s) and walks toward them along the graph, intercepting confusion mid-room before occupants reach a dangerous exit or jam a doorway.

- Static fire-unaware support can be harmful.

Signs and fixed guides consistently raised exposure - a concrete warning that fixed wayfinding should be paired with fire-aware, dynamically-updated guidance.

- ACO routing fails on complex maps.

Ant pheromone search does not converge on the multi-room topology (median ~476 s, many timeouts). RL-ACO therefore keeps A* for routing and reserves "optimization" for the decision-support layer.

Conclusion

Under limited visibility (50% unfamiliar) and a dynamic multi-source fire, an A*-based router augmented with mobile, hazard-intercepting guides achieves a robust trimmed-median T95 of 125.3 s (vs 205.9 s Dijkstra, 255.5 s A*, 476.1 s ACO), with the lowest exposure and fewest trapped. The gain is specifically attributable to mobility + proactive hazard interception - not to having guides per se, nor to where stationary guides are placed.

Future work: more fire scenarios and occupancies; leader-follower/communication among guides; learning the interception policy with reinforcement learning; seeded runs (common random numbers) to tighten variance; 3-D / multi-floor extensions.

Reproducibility

Source & simulator: github.com/feather0000/Evacuation_Program_RLACO * Report & data: github.com/feather0000/Evacuation_Report. Reproduce: `node tools/viz.js 12 2` (7 variants x 12 runs x 2 batches + 1 replay -> `js/viz-data.js`); `node tools/confirm.js 2` (cross-validation).